PRIVACY-PRESERVING RETRIEVAL-AUGMENTED GENERATION FOR TURKISH CONSUMER LAW: EVALUATION OF LOCALLY DEPLOYABLE LANGUAGE MODELS

SENIOR PROJECT REPORT
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Declaration

I hereby declare that this report is my own work, with the exception of the retrieval-augmented generation pipeline implementation and the underlying question-answer dataset, both of which were developed jointly as part of a senior project at TED University with the team members named in the Acknowledgments. All other content—including the evaluation framework, model comparisons, analysis, and the writing of this report—is solely my own. Passages and figures quoted from other works have been marked with appropriate mention of the source.

Place, Date	Signature

Abstract

The goal of this report is to explore whether local deployment of open-source language models, combined with a retrieval-augmented generation pipeline specialised to the domain, can reliably serve legal information on consumer protection law in Turkey. Applications of language models in the legal domain suffer from hallucination rates ranging between 17% and 88% depending on whether commercial retrieval-augmented models or closed-book frontier models are considered. In addition, the pertinent Turkish statutory framework is expressed in legal terminology that is not easily understood by non-specialists. We introduce HukukPusulası, a quality-aware retrieval-augmented generation system that leverages a domain-specialised knowledge base encompassing Law No. 6502, its 28 implementing regulations, and 372 judgments rendered by the Supreme Court of Turkey, alongside a retrieval module based on chunking, concept-based query enrichment, and quality-aware filtering of retrieved documents. From this corpus, we curated a stratified benchmark of 200 questions and performed, to the best of our knowledge, the first systematic evaluation of seven language models using the RAGAS framework against a locally deployed Llama-3.1-8B judge. The retrieval module consistently delivers retrieved evidence to all generators with an average Context Precision of 0.934 and Context Recall of 0.828. In the generation phase, Qwen-3-7B shows the best performance, significantly outperforming other models on Faithfulness and Factual Correctness and exceeding the proprietary Gemini-3.1-Flash-Lite baseline by 0.059 in Faithfulness (0.605 vs. 0.546) and 0.024 in Factual Correctness (0.564 vs. 0.540). These findings provide preliminary evidence that open-source models can be used to reliably answer consumer-law questions in Turkish using consumer-grade hardware.

Acknowledgments

Foremost, I would like to thank my mentor, Dr. Eren Ulu from the Computer Engineering Department of TED University, for suggesting this particular research topic to me, for the support I have received from him when the direction of this work was not yet clear, and finally, for the valuable and constructive comments during the process. Thanks to his willingness to give advice and his critical input on every single step of this project, I managed to greatly improve both the methods and the way of presenting this report.

This report is based on a senior project done together at TED University, and I would like to thank my fellow team members who contributed to the collective execution of the retrieval-augmented generation pipeline as well as the creation of the dataset. Everything else, including the evaluation scheme, comparisons of language models, analysing results, and the writing of this report, is my own.

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1 Introduction

1.1 Background and Motivation

Legal information access is an integral part of justice, but legal information is often inaccessible to people because of high prices for expert consultations and complex legal language. This situation manifests itself prominently within consumer-law in Turkey, as people encounter problems in their lives such as distance contract disputes, defective product issues, exercise of their withdrawal rights disputes, and warranty disputes, which are based on a body of law consisting of Law No. 6502 on Consumer Protection and its twenty-eight implementing regulations written in formal legal Turkish. Although large language models appear to solve the problem of generating natural responses, legal applications of language models have been shown to produce errors, or legal fictions, between 17% in commercial retrieval-augmented systems (Magesh et al., 2025) to 88% in closed-book frontier models (Dahl et al., 2024). Therefore, developing a trustworthy legal information resource based on large language models requires not only a technical solution but also a carefully designed methodology.

1.2 Problem Definition

This leads to the emergence of three interlinked challenges. Firstly, the system needs to refrain from producing false or misleading legal assertions, which necessitates the use of statutory text in retrievable form, instead of depending only on the parametric memory of the model. Secondly, the system must operate in Turkish, which is morphologically complex and therefore poorly supported by today's NLP infrastructure designed predominantly for English. Thirdly, in some legitimate use cases where the assistance of such a system is needed, notably including entities subject to the KVKK, the system should operate locally without using cloud APIs due to data privacy concerns, hence limiting the choice of architectures to open-source language models that can be hosted on consumer-grade hardware.

1.3 Research Questions and Objectives

This report aims to answer four research questions.

RQ1. What approaches can be used to measure the quality of the answers provided by language models to Turkish consumer-law questions in the absence of a benchmark for such questions?

RQ2. What is the performance of seven current language models, whose parameter size varies from 2B to 8B, on the four generation-phase metrics in answering Turkish consumer-law questions when provided with the same retrieval infrastructure?

RQ3. Can locally deployable open-source language models compete favourably against cloud-based proprietary language models, and if so, on which metrics?

RQ4. Does the retrieval framework that supplies input to the evaluated language models—that is, document-aware chunking, concept-based query enrichment, and quality-aware ranking—consistently produce high-quality evidence for all generators?

1.4 Contributions

The following three major contributions have been made by this report. Firstly, we construct a knowledge base in the domain of Turkish consumer law consisting of Law No. 6502, its twenty-eight implementing regulations, and 372 carefully selected judicial decisions from the Turkish Supreme Court (Yargıtay) and Regional Courts of Appeal. Secondly, we design and implement a quality-aware retrieval-augmented generation system which takes advantage of the structural and linguistic properties of Turkish legal documents, specifically document type-sensitive chunking consistent with the legislative article and clause structure, a concept-based query reformulation strategy which reduces the lexical gap between natural language consumer questions and legal terminology, and a distance-based ranking step which filters retrieved evidence prior to generation. Thirdly, we create a question-answer corpus from this knowledge base and conduct, to the best of our knowledge, the first systematic evaluation of seven state-of-the-art language models, namely, six open-source local models and one commercial cloud model, for Turkish consumer-law question answering using the RAGAS framework and a local judge model.

1.5 Report Organization

The structure of the report will be as follows. In Chapter 2, we will discuss the relevant literature on retrieval-augmented generation, evaluation criteria for these kinds of models, applications of language models to legal practice, and Turkish natural language processing. Chapter 3 will describe how the system that we propose has been designed, including the knowledge base, retrieval process, benchmark data set, and evaluation method. Chapter 4 will discuss experimental design and implementation. The experiments' results for all seven language models and six evaluation metrics will be discussed in Chapter 5. Results and their implications will be analysed in Chapter 6. The limitations of this study will be presented in Chapter 7, and Chapter 8 will be reserved for conclusions and future research directions.

2 Related Work

2.1 Retrieval-Augmented Generation

While language models pre-trained on large internet corpora have been shown to output fluent text when performing various open-ended tasks, the limitations of their knowledge-based abilities can be found in the frozen nature of their parameterised memory, which requires expensive training to be updated. If the query is located in the vicinity or even beyond the scope of such memory, the model will often output coherent yet unsupported text. In the case of legal applications, this flaw is particularly harmful. Dahl et al. (2024) document hallucination rates between 58% and 88% on US legal questions for closed-book frontier models. The picture barely improves once retrieval enters the loop: Magesh et al. (2025) measure rates of 17% to 33% for commercial retrieval-augmented legal assistants.

To address these limitations, Lewis et al. (2020) proposed the Retrieval-Augmented Generation (RAG) framework, which pairs a parametric generator with a non-parametric document index. During inference, a limited number of relevant passages are retrieved, and the generator is conditioned on the concatenation of the input question with the retrieved evidence. Updating the knowledge base requires only re-indexing without the need for retraining, a practical requirement for legal applications, where statutes and judicial precedents change constantly. Two retrieval families are common in this setting: sparse approaches like BM25 (Robertson and Zaragoza, 2009) and dense approaches like Dense Passage Retrieval (Karpukhin et al., 2020). In our work, we employ a dense retriever trained on Turkish texts, the choice of which we motivate in Section 2.4.

As RAG research expanded quickly, Gao et al. (2023) came up with a three-type taxonomy. The first one is Naive RAG, which is just the traditional retrieve-then-read pipeline. The second type is Advanced RAG, where extra processes can be inserted before or after retrieval. The third type is Modular RAG, in which the processes of retrieval and generation become interchangeable modules that can be either rearranged or omitted according to the input. Self-RAG (Asai et al., 2024) is an example of Modular RAG, which allows the generator to choose whether to retrieve via reflection tokens. Our system belongs to Advanced RAG with modular elements, namely concept-based query enrichment and quality-aware ranking.

Our approach builds directly on two recent studies. Mallen et al. (2023) demonstrate that retrieval gains are strongest for long-tail facts, exactly the regime Turkish legal questions occupy. Shi et al. (2024) argue that even when retrieval is performed, small generators tend to rely on their parametric prior rather than the retrieved evidence. With the help of strong retrieval, small generation models can produce results comparable to those of larger models, provided the retrieval phase delivers high-quality chunks. Our quality-aware ranking strategy is similar in spirit. By discarding low-quality retrieved chunks, we ensure that enough useful information is provided to the generator.

Our reasoning about model scale is supported by Borgeaud et al. (2022): a generator with 7.5 billion parameters, when augmented with retrieval, is comparable to GPT-3, with roughly 25 times more parameters. This finding clearly supports the assumption that we hypothesised—that an open-source model with 2 to 8 billion parameters can compete with a closed-source model in a specialised domain such as Turkish consumer law.

2.2 RAG Evaluation Frameworks

A human evaluation of RAG results cannot realistically be performed on a large scale when comparing various models on hundreds of queries, and it is likely to be inconsistent due to disagreements among raters. To address this problem, several evaluation approaches have been developed in the past three years, each involving a specific set of metrics.

Of these frameworks, the most commonly used one is RAGAS (Es et al., 2024). The RAGAS framework decomposes RAG quality into three elements that can be evaluated independently of any references: *Faithfulness*, which determines how grounded the generated answer is in the retrieved context, *Answer Relevancy*, which evaluates whether the answer corresponds to the input question, and *Context Relevance*, which determines how relevant the retrieved contexts are to the question. More advanced versions of the framework have also been developed, with additional metrics—such as Semantic Similarity and Factual Correctness—that evaluate answers relative to references. We use version 0.4.3 of RAGAS, where four reference-based and two reference-free metrics are combined.

Almost all these criteria depend upon the LLM-as-judge approach, where a language model is prompted to grade the generated responses according to a specific criterion. Zheng et al. (2023) established the empirical basis for this approach with MT-Bench, where agreement between GPT-4's preference and human judgment exceeded 80%. On the other hand, Wang et al. (2024) found that LLM judges show considerable bias towards position, verbosity, and self-enhancement, where the latter is particularly pronounced when the judge model is similar to the evaluated model. Two implications can be drawn from this discussion: care must be taken when interpreting results if the judge model is similar to one of the evaluated systems (we return to this self-enhancement concern in Chapter 7), and the generation parameters should be the same for all evaluated models.

Even the choice of a judge model comes with its own set of advantages and disadvantages. Using a cloud-based model like GPT-4 would entail costs as well as dependency on a third-party resource, not to mention that there might be rate-limit issues when many requests are made at once. Saad-Falcon et al. (2024) present ARES, a framework that shows how small, open-source models can match the performance of large, proprietary models in terms of human-annotator agreement if fine-tuned or prompted on in-domain synthetic data. Following the same approach, we chose Llama-3.1-8B as our judge model. Additionally, this choice corresponds to our use cases for the system, which do not involve any cloud resources and can be run completely offline.

Among the measures adopted by RAGAS, Factual Correctness is notable for its use in the legal domain. Min et al. (2023) proposed FActScore, which relies on breaking down the generated response into individual facts that can be independently verified against a reference answer. Such a breakdown is ideal for legal answers because they usually contain multiple statements (e.g., there is a withdrawal right, there is a fourteen-day period, and some product categories are exempt). An overall similarity score between a generated answer and the reference may fail to distinguish a fully correct response from one in which almost all facts are correct but a single statement is wrong.

As mentioned above, these models assessing retrieval and generation have been developed for benchmarking purposes for the English language. Hence, there is not enough literature regarding the performance of these models on morphologically rich languages such as Turkish or on low-resource languages. Particularly, there is a lack of research in the domain of law. This study aims to bridge this gap by testing the models on Turkish legal documents.

2.3 Legal AI Systems

Legal applications of language models have been extensively explored in the last few years, from showcasing the abilities of the latest models in answering legal exams to benchmarking their legal reasoning capabilities. Nonetheless, there are two key differences between legal AI and other knowledge-intensive AI systems that have influenced the direction of recent work in this field: an exceptionally high rate of factual hallucinations and the limited support for low-resource languages.

Firstly, the problem of factual hallucinations is well-documented. In a study conducted by Dahl et al. (2024), multiple frontier closed-book language models were tested on US legal questions and were found to generate hallucinations at rates ranging from 58% for GPT-4 to 88% for Llama 2. The implications of these results extend beyond the numbers themselves, since they clearly illustrate that the field of law imposes requirements for factual correctness, which parametric models cannot meet regardless of their scale. Fabricated case law and statutory citations can be especially detrimental in legal applications since end users will take the quoted sources at face value without further checking.

A natural reaction to this problem has been to complement legal AI with retrieval-based systems. However, according to Magesh et al. (2025), commercial retrieval-augmented legal assistants such as Lexis+ AI, Westlaw AI-AR, and even GPT-4 fine-tuned with bespoke RAG pipelines hallucinate between 17% and 33% of the time on common legal queries. The fact that RAG can reduce but not solve the legal hallucination problem provides one of the main motivations for a key methodological decision taken in this work: introducing a quality-based post-retrieval ranking step, described in Chapter 3, prior to generation.

The second problem, namely language representation, is apparent in the prominent benchmarks of legal AI. LegalBench (Guha et al., 2023), which consists of 162 collaboratively developed legal reasoning tasks, covers only English. Another English-only resource, LexGLUE (Chalkidis et al., 2022), brings together seven datasets from the European

Court of Human Rights, US Supreme Court, and EU legislation. Even the multilingual benchmark LEXTREME (Niklaus et al., 2023) and its companion dataset MultiLegalPile, which together cover 24 languages, omit Turkish. Commercially available legal AI tools, such as Harvey, Lexis+ AI, Westlaw AI-AR, and CoCounsel, are also closed-source, English-language, and priced for enterprise law firms. To the best of our knowledge, no peer-reviewed Turkish consumer-law benchmark or RAG system currently exists.

Both of these factors combined—the continuing hallucinations of legal AI and the absence of Turkish in its benchmarking instruments—form the empirical and methodological context of this report. In the following section, we will consider the current state of affairs in the Turkish NLP area, which serves as the necessary technological basis for any Turkish consumer-law framework.

2.4 Turkish NLP and Legal Domain

The properties of the Turkish language pose some challenges, which make Turkish different from the English-language settings in which the majority of current NLP efforts are developed. Since Turkish is an agglutinative language, a word is created by adding multiple suffixes to the stem, leading to the situation where one English word can be represented in Turkish by several surface forms (tüketici, tüketiciye, tüketicinin, tüketicilerden). This imbalance becomes concrete in tokenizer behaviour. Petrov et al. (2023) find that subword segmentation models trained on English-heavy corpora produce two to four times more tokens for Turkish than for English text of equal semantic content. Both the context window and the inference cost suffer as a result. Toraman et al. (2023) confirm this effect across six downstream Turkish NLP tasks and show that a Turkish-specific tokenization approach is crucial for information retrieval and classification.

One way of addressing the issues listed above is the creation of Turkish-specific encoders. Schweter (2020) introduced BERTurk as a collection of cased and uncased BERT models pre-trained on about 4.4 billion Turkish tokens; BERTurk has become the default encoder for Turkish NLP. Subsequently, Uludoğan et al. (2024) introduced TURNA, which is a Turkish encoder-decoder model based on the UL2 architecture, and found that monolingual Turkish models significantly outperformed multilingual models on Turkish datasets. This finding bears great significance for the design of our retrieval: any multilingual sentence encoder, regardless of its size, is likely to underperform a smaller Turkish-specific encoder. Our retrieval system thus uses a Sentence-BERT variant based on BERTurk.

Turkish legal NLP is a young field with limited published work. The first named-entity recognition model for Turkish judicial documents comes from Çetindağ et al. (2023). Their model uses a character-level BiLSTM with GloVe and Morph2Vec embeddings, achieving an F1 of 92.27% on five entity types, namely citations, article IDs, chamber names, litigant names, and dates. BERTurk-Legal, released by Öztürk et al. (2023), adapts BERTurk to the legal domain through fine-tuning on Yargıtay court decisions. The intended downstream task is previous-case retrieval, not question answering. Both works highlight the value of domain adaptation for Turkish legal documents, but neither addresses retrieval-augmented generation or consumer law.

In light of this context, it is possible to articulate the contribution made by the current report. To the best of our knowledge, there is no published academic study investigating Turkish consumer law via retrieval-augmented generation, nor is there any academic paper conducting a systematic analysis of open-source language models for answering legal questions in Turkish. None of the multilingual legal benchmark datasets presented in Section 2.3 contains Turkish legal text, and the Turkish legal NLP studies reviewed above have focused on classification and information retrieval tasks on case law rather than legislative statutes. Furthermore, consumer law, which has been primarily codified in Law No. 6502 and its twenty-eight implementing regulations, has yet to be systematically investigated in the context of NLP. This report seeks to bridge this research gap by building a specialised knowledge corpus based on statutes and case law, developing a quality-aware retrieval pipeline for Turkish legal documents, and providing the first systematic analysis of seven language models on a constructed benchmark dataset.

3 Methodology

3.1 System Architecture Overview

HukukPusulası is a domain-focused retrieval and generation system built on the RAG architecture for Turkish consumer law. Figure 1 shows the overall design. The pipeline has four stages: document collection from Turkish legal resources (mevzuat, yargıtay karar arama), vectorization, retrieval with quality tiering, and generation by large language models. Three design choices distinguish this system from a generic RAG pipeline. The chunking module is document-aware: it splits Turkish legal texts along natural boundaries such as articles, clauses, court decisions, and legislative acts, while handling Turkish morphology. We also enrich queries with scenario-specific legal concepts before retrieval, narrowing the gap between everyday Turkish and formal legal language. After retrieval, a quality-tiering filter drops low-confidence chunks before they reach the generator, in order to reduce the risk of hallucination.

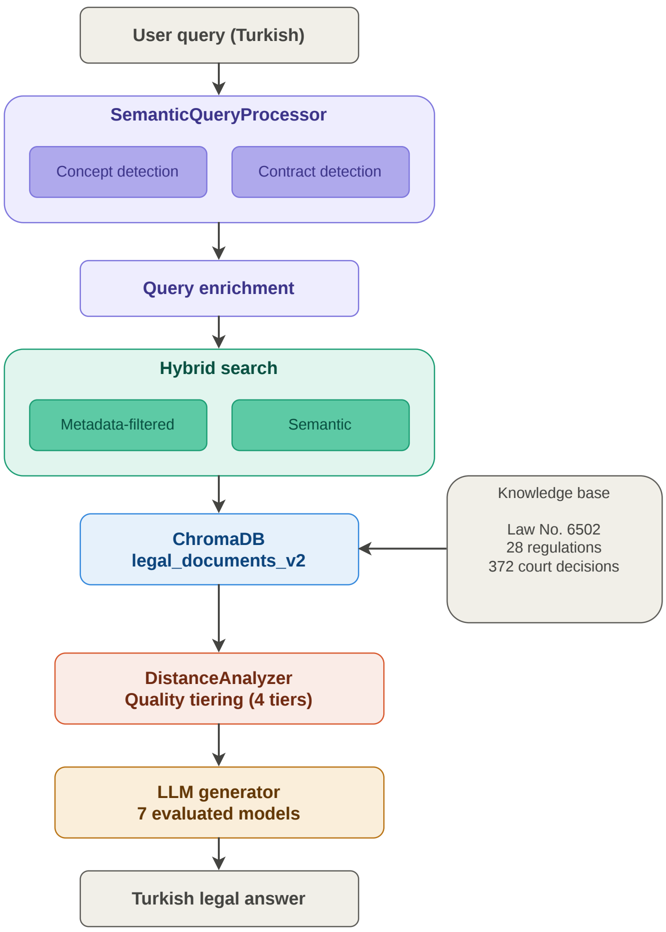

Figure 1: Overall architecture of the HukukPusulası system. A Turkish consumer query is processed through concept-based query enrichment, retrieved via hybrid search over a vector store of Turkish legal documents, filtered by quality tiering, and answered by an LLM generator.

3.2 Legal Knowledge Base Construction

3.2.1 Source Documents

The legal corpus consists of three types of documents that serve as authorities in Turkish consumer law (Table 1):

- **Primary Legislation:** Turkish Consumer Protection Law (Law No. 6502), which forms the legislative foundation of consumer protection in Turkey.
- **Implementing Regulations:** 28 regulations have been adopted based on the Consumer Protection Law and relate to topics such as distance selling, hire purchase, contracts away from premises, consumer credit, mortgages, advertising, guarantees, and consumer disputes settlement boards.
- **Court Decisions:** 372 rulings from the Turkish Supreme Court (Yargıtay) and regional Courts of Appeal.

All documents were collected and retrieved in their official PDF form and preprocessed through the PyMuPDF library for the text extraction phase.

Table 1: Composition of the legal knowledge base.

Document Type	**Count**	**Share (%)**
Primary Legislation (Law No. 6502)	1	0.25
Implementing Regulations	28	6.98
Court Decisions (Yargıtay)	372	92.77
Total	**401**	**100.00**

3.2.2 Document-Aware Chunking

We developed the LegalDocumentChunker function that processes all document types used in our corpus (Kanun [laws], Yönetmelik [regulations], and Yargıtay Kararları [Court of Cassation decisions]). For laws and regulations, the function performs segmentation based on articles, which can be identified by the structure MADDE X (Article X). These articles are further divided into paragraphs and sub-clauses in the format m.15/1-a (Article 15, paragraph 1, sub-clause a). In the case of court decisions, every decision is processed as one chunk, thus maintaining the context of the entire case. The chunker parameters include a maximum chunk size of 3000 characters, an overlap of 300 characters, and a minimum chunk size of 500 characters. These size limits prevent chunks from carrying either insufficient or excessive information. Lastly, each chunk also holds structural metadata—document type, source, and article number—allowing selective filtering when retrieving. The current metadata structure is deliberately simple; extensions are addressed in Chapter 7.

3.2.3 Embedding Model

For vectorization, we use `emrecan/bert-base-turkish-cased-mean-nli-stsb-tr`, which is the Turkish version of BERT fine-tuned on NLI and STS tasks. This model serves our purposes for two main reasons. First, the model accounts for the structural and lexical properties of the Turkish language, which include legal terms that might not be captured by multilingual general-purpose models. Second, fine-tuning on NLI/STS results in high-quality sentence-level semantic embeddings, which are useful in the next phase. While we cannot claim this to be the optimal choice, the use of a Turkish-tailored encoder allows us to minimise language as a confounding variable so that the comparison across generation models reflects generator quality rather than encoder quality. The model generates 768-dimensional embeddings with a maximum sequence length of 512 tokens (roughly 2000–3000 characters for Turkish text). The chunking parameters—minimum chunk size, maximum chunk size, and overlap—were chosen based on the constraints of this embedding model. During preprocessing, we use the model's BERT tokenizer, handling overflow by chunking rather than truncating.

3.2.4 Vector Store

Our vector database is ChromaDB in persistent mode with cosine similarity as the distance metric. The collection we have defined in ChromaDB is called `legal_documents_v2`. Our collection includes the embedded chunks together with their metadata consisting of document type (`laws`, `regulation`, `court_decision`, `generic`), article number, and sub-clause information. This will help us improve retrieval strategies and enable further extensions to the system.

3.3 Specialised Retrieval Pipeline

The main novel contribution of our work lies in the design of a query processing pipeline which is based on the morphological properties of the Turkish language and Turkish consumer-law terminology. We make two crucial decisions, one being choosing the right kind of embedding model which was explained in the embedding model section, while the other is choosing the architecture of our pipeline. Our pipeline can be divided into three sections; namely concept-based query enrichment, hybrid search, and quality ranking.

3.3.1 Concept-Based Query Enrichment

Most consumer queries about legal matters come in everyday Turkish. The underlying legal texts do not. They rely on formal vocabulary that demands specialised training. To bridge this gap, we built the `SemanticQueryProcessor`, which matches each user input against precomputed concept embeddings.

In the first detection phase, the module identifies four different legal concepts based on legal queries, namely cayma hakkı, ayıplı mal, garanti, and tazminat. For each concept, we defined a few example queries that consumers use in their inputs. We then embed these examples using the Turkish BERT model, which captures formal legal terminology more accurately than multilingual alternatives, and compute the concept embedding as the centroid of the example embeddings.

The second phase starts with the type of consumer agreement mentioned in the search request: mesafeli sözleşmeler (distance agreements, commonly used in e-commerce transactions), iş yeri dışında kurulan sözleşmeler (agreements made outside business premises, such as door-to-door sales), and taksitli satış sözleşmeleri (instalment sales agreements). Each of the agreements listed above corresponds to a chapter of the Turkish Consumer Protection Law with different implementing regulations. The detection method we chose to implement in our pipeline works using nearly the same mechanism as the embedding similarity method, with a threshold of approximately 0.45. After choosing concepts, the enriched version of the query is created and used for vector-based search.

3.3.2 Hybrid Search

There are two retrieval channels operating in sequence. First, upon detecting the contract type, the system performs a *metadata-filtered search* within the pool of documents related to that contract type. Then a *semantic search* is conducted using the enriched query over the entire corpus. The results of both searches are combined, with duplicate removal based on document hashing.

3.3.3 Quality-Aware Ranking

Even with a strong retriever, irrelevant chunks sometimes slip through and add noise during the LLM generation process. We address this with the `DistanceAnalyzer` component. In its full design, the `DistanceAnalyzer` sorts each retrieved document into one of four quality bands (Excellent, Good, Acceptable, Poor), based on its cosine distance to the query. The experiments reported in this report use a simpler form of this component: a single cut-off at 0.45 cosine distance. Chunks above the cut-off are dropped, and the rest are passed to the generator. Calibrating the full four-band scheme on our benchmark is left as future work and discussed in Chapter 7.

3.4 Worked Example: Tracing a Consumer Query

To show how the pipeline activates on a real query, we walk through an example that we provide below.

Input Query. The user's question in Turkish is:

Query

"Tüketici Hakem Heyeti tam olarak nedir ve ne işe yarar?"
("What exactly is the Consumer Arbitration Committee (THH) and what does it do?")

Concept and Contract Detection. The SemanticQueryProcessor matches the input with one of the items we listed (cayma hakkı, ayıplı mal, garanti, tazminat) and the three types of contracts. Since this sample query relates to the arbitration committee, the input continues on its path. Furthermore, the specific article from which our ground truth was created is not retrieved, but there are several others that explain the question asked by the user.

Hybrid Search. A semantic search runs on the entire corpus, returning 5 chunks, all of which come from the Tüketici Hakem Heyetleri Yönetmeliği (Regulation of Consumer Arbitration Committees):

- Article 2 — scope of the regulation
- Article 5 — establishment of committees by the Ministry
- Article 8 — committee composition (chair and four members)
- Article 10 — consumer-organisation representative
- Article 20 — binding nature of committee decisions

Quality Tiering. The `DistanceAnalyzer` measures the cosine distance of each retrieved chunk to the enriched query. Following the simplified scheme used in our experiments (Section 3.3), chunks above the 0.45 cut-off are dropped, and the rest are passed to the generator.

Answer Generation. The chosen language model in the example we provide (Gemma-3-4B) gets the input query together with the filtered chunks and generates the following output (an excerpt):

Model Output (Gemma-3-4B)

"Tüketici Hakem Heyetleri (THH), tüketiciler ve satıcılar arasındaki anlaşmazlıkları arabuluculuk yoluyla çözmelerine yardımcı olan kuruluşlardır. THH'lerin temel amacı, tüketicilerin karşılaştığı maddi ve manevi zararların arabuluculuk yöntemiyle çözülmesini sağlamaktır..."

The output is clear for those not specialised in the legal domain. It is simple to read and it is written in a daily conversation style. Furthermore, this model has the highest semantic similarity. Conversely, it is not the best answer we gathered.

3.5 Benchmark Dataset Construction

3.5.1 Q&A Generation Pipeline

In order to evaluate the performance of LLMs on Turkish consumer law, we created a benchmark by generating question-answer pairs from documents. For this purpose, we used Gemini-2.5-flash-lite, a cloud-based LLM accessible via the Google AI Studio API, as our generation model. Our generation model operates at the sub-clause level, addressing the structural format of Turkish legal documents (e.g., m.15/1-a). Specifically, we generated approximately three question-answer pairs per sub-clause for laws and regulations and five pairs per court decision.

Three quality criteria shape the Q&A generation prompt. The first is plain, accessible language: questions must be readable without legal training. The second is content coverage, with required attention to exceptions, merchant rights, procedural restrictions, and consumer rights. The third is reasoning variety, with three question types targeted: hypothetical situations, analytical questions, and factual questions. We chose this distribution to test different reasoning skills. Furthermore, generated answers are in a non-legal, conversational style accessible to any user, especially those unfamiliar with legal terms.

In total, we obtained 13,676 Q&A pairs by running the prompt over one main law, 28 regulations, and 372 court rulings. The prompt also requires a specific JSON format for each output. Moreover, 588 out-of-domain rejection patterns were created to enable the system to reject queries that fall outside the scope of consumer law.

3.5.2 Benchmark Subset

To assess the performance of language models, we selected a sample of 200 questions through stratified sampling. This way, the questions are balanced in terms of the number of source documents and question types. We evaluated all 200 questions across all seven language models, yielding 1,400 evaluation cases in total. The test sample contains the following proportions of question types: hypothetical situations (50%), analytical questions (30%), and factual questions (20%).

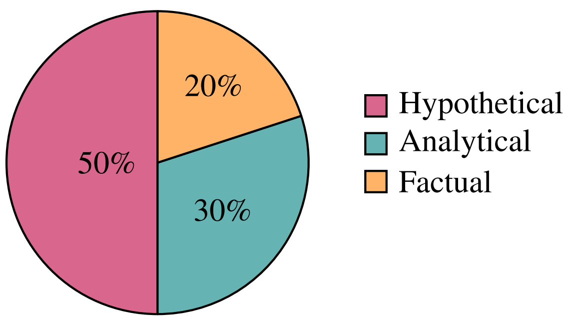

Figure 2: Distribution of question types in the 200-question benchmark subset.

3.6 Evaluated Language Models

Seven large language models with different sizes and architectures were selected for the analysis. Six of these models are open-source and tested locally through Ollama (v0.23.3), while the seventh model is a proprietary cloud-based baseline model. Further details about the selected models are provided below:

Table 2: Evaluated language models.

Model	Params	Type	Provider
Gemma-2-2B	2B	Local	Google
Gemma-3-4B	4B	Local	Google
Llama-3.1-8B	8B	Local	Meta
Mistral-7B	7B	Local	Mistral AI
Qwen-3-7B	7B	Local	Alibaba
DeepSeek-R1-Distill-Qwen-7B	7B	Local	DeepSeek
Gemini-3.1-Flash-Lite	—	Cloud	Google

The model selection criteria were the following: (1) various model families reflecting architectural diversity, (2) model sizes ranging from 2B to 8B parameters, fitting the hardware limitations of our deployment environment, which is restricted to 8 GB VRAM, (3) permissive licensing for open-source models, and (4) the inclusion of both reasoning-specialised models (DeepSeek-R1) and general-purpose models, as our goal was to investigate whether reasoning abilities generalise to other domains. Generation parameters were the same for all local models provided by Ollama (`temperature=0.0`, `context length=8192`), whereas Gemini-3.1-Flash-Lite was accessed via Google's free public API with the same parameters.

3.7 Evaluation Framework

3.7.1 RAGAS Metrics

The RAGAS metric suite (v0.4.3) is used for systematic assessment, and six related metrics covering various aspects of RAG system performance are calculated:

- **Context Precision (with Reference):** measures whether the retrieved context snippets are relevant to the correct answer.
- **Context Recall:** checks whether the provided context contains all the facts needed to produce a correct answer.
- **Faithfulness:** scores how well the generated response is grounded in the retrieved context; a low score signals hallucinations.
- **Answer Relevancy:** indicates how closely the generated answer addresses the input question.

- **Semantic Similarity:** computes embedding-based cosine similarity between the generated and reference answers.
- **Factual Correctness (F1 mode):** aligns atomic facts of the generated and reference responses to produce an F1-style score.

Of these six, Context Precision, Context Recall, Faithfulness, Answer Relevancy, and Factual Correctness rely on an LLM-as-a-judge setup, while Semantic Similarity uses embeddings directly.

3.7.2 Judge Model Selection

We chose `Llama-3.1-8B` as the judge model for three main reasons. Firstly, Llama-3.1-8B provides a good trade-off between evaluation performance and the computational resources needed to deploy it in our setting (i.e., 8 GB VRAM). Secondly, using a cloud-based judge like GPT-4 would impose costs related to API access and would create a dependency that goes against the privacy-oriented architecture of our system. Lastly, using a free and open-source judge improves reproducibility, allowing other researchers to reproduce our work even if they lack access to any commercial APIs. The selection of the judge model may lead to systematic bias, which we will further discuss in Chapter 7.

3.7.3 Embedding Model for Semantic Similarity

For Semantic Similarity and Answer Relevancy, we compute embeddings using `nomic-embed-text`, an open-source multilingual model accessed through Ollama. This keeps the system consistent with our open-source-first approach. Table 3 summarises all pipeline hyperparameters.

4 Experiments

This chapter describes the experimental setup of the study. It basically provides information about the local LLM systems that we implemented and evaluated, the configuration used in the evaluation phases described in Chapter 3, the experimental procedure, the way we chose to evaluate our systems, and the benchmarking subset.

4.1 Hardware and Software Environment

Experiments were conducted locally within a single computer to safeguard the user's privacy regarding their queries, which motivates the present study. Such a design avoids recording any important data to the database accessible by the cloud servers, for example, API endpoints such as Google or ChatGPT. The hardware consists of an Intel Core i7-13620H processor (13th generation, hybrid core layout with both performance and efficiency cores), 16 GB of RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB of GDDR6 video memory, used for local language model inference. The operating system is Windows 11 Home. The software for local language model inference uses Ollama 0.23.3, a free, open-source engine that provides a uniform HTTP endpoint to quantised language models stored in the GGUF file format and dynamically allocates the load of model inference between the available CPU and GPU hardware. The retrieval pipeline, evaluation, and other supporting code, such as scraping documents from legal texts (mevzuat), are written in Python 3.13.2. The main third-party packages used include RAGAS 0.4.3 for evaluation, LangChain together with the `langchain-ollama` package for model orchestration, ChromaDB configured in persistent mode as the vector index, and `sentence-transformers` for generating embeddings. The full hardware and software configuration alongside the pipeline hyperparameters is given in Table 3.

Table 3: Key hyperparameters and configuration values used throughout the HukukPusulası pipeline.

Component	Parameter	Value
Chunking	Maximum chunk size	3,000 characters
	Minimum chunk size	500 characters
	Overlap	300 characters
Embedding	Model	`emrecan/bert-base-turkish-cased-mean-nli-stsb-tr`
	Dimension	768
	Max sequence length	512 tokens
Query Processing	Similarity threshold	0.45
	Legal concepts	4
	Contract types	3
Vector Store	Database	ChromaDB
	Distance metric	Cosine
	Collection	`legal_documents_v2`
Quality Ranking	Cosine-distance cut-off	0.45
	Quality bands (designed)	4 (Excellent/Good/Acceptable/Poor)
Generation	Temperature	0.0
	Context window	8,192 tokens
Evaluation	Judge model	Llama-3.1-8B
	Embedding model (eval)	`nomic-embed-text`
	Timeout per case	600 seconds
	Retries on failure	3
	Cool-down	60s per 10 evaluations

Consumer-grade hardware was chosen on purpose. We tested 2 to 8 billion parameter models, paired with a focused pipeline, to see if they can handle Turkish consumer law questions without needing cloud LLMs or industry-grade hardware. However, this only works if the user's system is capable enough. Table 4 lists the hardware and software.

Table 4: Hardware and software environment used for the experiments.

Component	Specification
CPU	Intel Core i7-13620H (13th gen)
RAM	16 GB
GPU	NVIDIA GeForce RTX 4070 Laptop (8 GB GDDR6)
Operating system	Windows 11 Home
LLM runtime	Ollama 0.23.3
Python	3.13.2
Evaluation	RAGAS 0.4.3
Vector store	ChromaDB (persistent mode)
Embedding library	`sentence-transformers`
Embedding model	`emrecan/bert-base-turkish-cased-mean-nli-stsb-tr`
Orchestration	`langchain-ollama`

4.2 Judge Model and Embedding Configuration

The calculation of the RAGAS metrics involves two supporting models, both running locally via Ollama. The judge model employed for the five LLM-as-judge metrics (Context Precision with Reference, Context Recall, Faithfulness, Answer Relevancy, and Factual Correctness) is Llama-3.1-8B with a temperature value of 0 and a context window size of 8192 tokens. The decision to employ an open-source judge model with 8 billion parameters is based on the following three rationales, as elaborated in Section 2.2: ensuring reproducibility for subsequent research efforts, avoiding the recurring expense of API calls that would otherwise limit the scope of evaluation, and conformity to the privacy-preserving deployment setting where the entire system needs to run locally. The second supporting model employed for the computation of the two metrics involving embeddings (Semantic Similarity and the embedding part of Answer Relevancy) is `nomic-embed-text`. Note that Llama-3.1-8B is also one of the seven models evaluated in this work; the potential bias arising from this overlap is addressed in Chapter 7.

4.3 Benchmark Subset and Reporting

We evaluate on a stratified sample of 200 questions, drawn from the 13,676 question-answer pairs described in Chapter 3. The sample was stratified to focus on three reasoning-intensive question types from the full dataset: approximately 50% hypothetical, 30% analytical, and 20% factual questions. Each of the seven models under review is tested on the entire 200-question sample, producing 1,400 output samples, and each sample is evaluated against six different metrics, totalling 8,400 metric observations. We run each evaluation deterministically, so the same input always produces the same output. The output data is saved for the analysis in Chapter 5.

5 Results

This chapter presents the results of the empirical evaluation process discussed in Chapters 3–4, following the two-stage evaluation approach proposed in the methodology: first, retrieval-phase performance; second, generation-phase performance across four metrics computed on the generator's output.

5.1 Retrieval-Phase Performance

Table 5 provides the Context Precision and Context Recall metrics for all (model, question) pairs tested. Since the retrieved contexts are identical for all seven models, the retrieval evaluation results can be averaged across models rather than reported for each model separately. Minor differences in performance, amounting to less than two percentage points and resulting from judge stochasticity, will be addressed in Chapter 6.

Table 5: Retrieval-phase performance averaged across all evaluations.

Metric	**Average Score**
Context Precision (with Reference)	0.934
Context Recall	0.828

5.2 Generation-Phase Performance

The results of the generation phase are presented in Table 6, where the highest scores for the four measures in each column are presented in bold font.

Table 6: Generation-phase performance per model. Best value in each column is shown in bold. † Refers to DeepSeek-R1-Distill-Qwen-7B (Ollama tag: `deepseek-r1:7b`).

Model	**Faithfulness**	**Ans. Relevancy**	**Sem. Similarity**	**Factual Corr.**
Qwen-3-7B	**0.605**	0.462	0.791	**0.564**
Gemini-3.1-Flash-Lite	0.546	0.455	0.791	0.540
Gemma-3-4B	0.535	0.454	**0.803**	0.521
Gemma-2-2B	0.326	0.404	0.787	0.390
Llama-3.1-8B	0.294	**0.478**	0.743	0.319
Mistral-7B	0.273	0.436	0.771	0.278
DeepSeek-R1-7B†	0.247	0.428	0.595	0.278

On the metrics of Faithfulness and Factual Correctness, which directly concern the legal grounding goal of the system, Qwen-3-7B obtains the best results (0.605 and 0.564, respectively). The proprietary cloud-based solution, Gemini-3.1-Flash-Lite, ranks second on both (0.546 and 0.540). For the other two metrics, the spread is less pronounced: Llama-3.1-8B obtains the highest result on Answer Relevancy (0.478), and Gemma-3-4B performs best on Semantic Similarity (0.803).

6 Discussion

In this chapter, we highlight three of these results and speculate on their meaning in terms of our larger research question about the feasibility of using locally deployable retrieval-augmented language models as a foundation for Turkish consumer-law information access.

First, we observe an impressive consistency of performance at the retrieval stage. Namely, the average values of Context Precision and Context Recall are 0.934 and 0.828 respectively. Crucially, what we see here is not the value itself, but the absence of any material variation across the seven models. As explained in Chapter 7, the reason is residual judge stochasticity and not the retrieval per se. In practical terms, this means that our retrieval module—consisting of document-aware chunking, query enrichment, and quality-aware ranking—works independently of the underlying generator and does not act as a bottleneck for downstream performance.

Second, and more significantly, the comparison between deployable open-source models and the proprietary cloud benchmark reveals that Qwen-3-7B outperforms Gemini-3.1-Flash-Lite by 0.059 in terms of Faithfulness (0.605 versus 0.546) and by 0.024 in terms of Factual Correctness (0.564 versus 0.540), the two metrics most immediately relevant to legal grounding, and performs essentially equivalently (within less than one percentage point) in the other two generation-phase metrics. This finding holds despite an apparent advantage that should favour Gemini because the ground-truth answer data used in the benchmark were generated by Gemini-2.5-Flash-Lite, a model belonging to the same model family as Gemini-3.1-Flash-Lite and thus likely to provide stylistic affinity in reference-based metrics. If an unrelated open-source model can still outperform the proprietary benchmark model on Faithfulness and Factual Correctness despite such a possible advantage, then it is reasonable to conclude that the advantage, assuming it exists at all, must be relatively minor. In any event, when deployment must satisfy Turkish data protection requirements or is otherwise restricted by API costs, this means that a deployable seven-billion-parameter open-source model is a realistic option for Turkish consumer-law question answering.

This third result is surprising and deserves more exploration. DeepSeek-R1-Distill-Qwen-7B is a reasoning model created by distilling the chain-of-thought behaviour of DeepSeek-R1 (a model trained with a reinforcement learning objective) into a Qwen-7B base, and attains the lowest Faithfulness score (0.247) and one of the lowest Factual Correctness scores (0.278) among the seven models studied. The opposite effect was expected from reasoning fine-tuning. This can be explained by the nature of the reinforcement learning objective which, through encouraging chain-of-thought behaviour, causes the model to base its responses on its own reasoning trace rather than the retrieved evidence; this conclusion is in line with the observation that motivated Context-Aware Decoding (Shi et al., 2024). Assuming this is the correct explanation, reasoning-fine-tuned models should not be expected to perform better within the retrieval-augmented framework.

The above-described results taken as a whole provide evidence for the technical hypothesis of this report: a domain-specific retrieval module coupled with a chosen generator of seven billion parameters available under an open-source license is capable of providing Turkish consumer-law question answering that matches the quality of cloud-based commercial competitors, running on consumer-grade hardware. While these results do not prove deployment feasibility, which would require a human-curated benchmark as outlined in Chapter 7, they do establish technical feasibility, upon which the social motivation of this report depends.

7 Limitations

First, the ground-truth answers used for benchmarking were created by the Gemini-2.5-Flash-Lite model and not verified by a human domain expert. Although the generation prompt required the answers to be grounded in the retrieved statutory text, any potential errors and biases in the generation process can also affect the output of the four reference-based metrics of our evaluation suite. Consequently, we interpret the four reference-based metrics alongside the two reference-free metrics. The highest priority for future improvement in this direction would be a verification of the benchmark answers performed by a practising Turkish consumer-law specialist.

Second, Llama-3.1-8B acts as the judge model and is one of the seven tested models. Such an overlap raises concerns about self-enhancement bias (Wang et al., 2024) among the results. However, Llama-3.1-8B does not show systematic dominance across the metrics: it ranks in the middle of the seven tested models on Faithfulness and Factual Correctness, which contradicts a strong presence of self-enhancement bias. Ideally, the design of such experiments would involve an independent model family or an ensemble of judges with disagreement calibration.

Third, there is a potential for stylistic bias in favour of the model family containing both the dataset generator (Gemini-2.5-Flash-Lite) and one of the tested models (Gemini-3.1-Flash-Lite). Nonetheless, Qwen-3-7B performs better than Gemini-3.1-Flash-Lite on the metrics most relevant to legal grounding (Faithfulness and Factual Correctness), which indicates that if such an effect exists, it has little influence.

Fourth, the number of questions included in the benchmark is about 1.5% of the total of 13,676 question-answer pairs in the entire corpus. Rankings among closely competing models may be sensitive to variations given the sample size. In addition, the report is limited to Turkish consumer law as provided under Law No. 6502 and further clarified in its implementing regulations, along with a static knowledge base that does not account for legislative changes or new legal precedents after the date of construction.

Fifth, our test hardware has only 8 GB of GPU memory. This caps locally runnable models at roughly 8 billion parameters in standard quantisation and rules out 13–14B versions. The cap is intentional: the report targets privacy-friendly deployment, which by definition needs consumer-grade hardware. Therefore, any hardware constraint can be matched with a suitable model. The main reason for choosing this topic is that privacy-critical fields, such as the legal domain in our case, can be served while using local models. Ultimately, this trade-off can be balanced.

Sixth, the `DistanceAnalyzer` component described in Section 3.3 was designed with a four-tier classification scheme (*Excellent*, *Good*, *Acceptable*, *Poor*) based on the cosine distance between the retrieved chunk and the input query. For the experiments reported in this report, however, we used a single-threshold simplification of this scheme at a cosine distance of 0.45, based on the test results we gathered. Calibrating and evaluating the full four-band classification on the constructed benchmark is left as a direction for future work.

8 Conclusion and Future Work

This report presents HukukPusulası, a retrieval-augmented generation system that integrates quality-aware mechanisms for the domain of Turkish consumer law, along with the evaluation of seven language models on a Turkish consumer-law benchmark. Our system uses laws, regulations, and court decisions as its knowledge base, along with an enriched retrieval component that applies document-aware chunking, concept-based query enrichment, and quality-aware ranking. The four research questions can be answered as follows.

RQ1: The measurement protocol uses six metrics: two for the retrieval phase (Context Precision and Context Recall) and four for the generation phase (Faithfulness, Answer Relevancy, Semantic Similarity, and Factual Correctness). Five rely on the Llama-3.1-8B judge; Semantic Similarity uses embeddings.

RQ2: On generation performance overall, Qwen-3-7B leads. It achieves the highest scores on Faithfulness and Factual Correctness, and performs near-equally on the other two metrics.

RQ3: Compared to Gemini-3.1-Flash-Lite, Qwen-3-7B matches it on Answer Relevancy and Semantic Similarity, while outperforming it on Faithfulness and Factual Correctness.

RQ4: All generators receive consistent retrieval results, with an average Context Precision of 0.934 and Context Recall of 0.828. The minor variation between models arises from the stochastic nature of the judge.

There are several possibilities for future research in light of the limitations pointed out in Chapter 7: developing a benchmark test based on human review by a practising lawyer specialising in Turkish consumer law, the most urgent possibility; employing an ensemble of judges drawn from independent families to mitigate self-enhancement and model-family bias; extending the scope of the corpus beyond consumer law to cover other branches of Turkish law such as criminal, family, or commercial law; and testing larger models of 13 to 14 billion parameters on server-class machines. All considered, these findings represent early indications that restricted-domain, restricted-language, locally-deployable legal AI is possible, an important step toward increasing the accessibility of Turkish consumer law for citizens currently unable to access it.

References

A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. URL `https://openreview.net/forum?id=hSyW5go0v8`.

S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, G. B. Van Den Driessche, J.-B. Lespiau, B. Damoc, A. Clark, D. De Las Casas, A. Guy, J. Menick, R. Ring, T. Hennigan, S. Huang, L. Maggiore, C. Jones, A. Cassirer, A. Brock, M. Paganini, G. Irving, O. Vinyals, S. Osindero, K. Simonyan, J. Rae, E. Elsen, and L. Sifre. Improving language models by retrieving from trillions of tokens. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu, and S. Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 2206–2240. PMLR, 17–23 Jul 2022. URL `https://proceedings.mlr.press/v162/borgeaud22a.html`.

C. Çetindağ, B. Yazıcıoğlu, and A. Koç. Named-entity recognition in turkish legal texts. *Natural Language Engineering*, 29(3):615–642, 2023. doi: 10.1017/S1351324922000316.

I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androutsopoulos, D. Katz, and N. Aletras. LexGLUE: A benchmark dataset for legal language understanding in English. In S. Muresan, P. Nakov, and A. Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4310–4330, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.297. URL `https://aclanthology.org/2022.acl-long.297/`.

M. Dahl, V. Magesh, M. Suzgun, and D. E. Ho. Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis*, 16(1):64–93, 01 2024. ISSN 2161-7201. doi: 10.1093/jla/laae003. URL `https://doi.org/10.1093/jla/laae003`.

S. Es, J. James, L. Espinosa Anke, and S. Schockaert. RAGAs: Automated evaluation of retrieval augmented generation. In N. Aletras and O. De Clercq, editors, *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158, St. Julians, Malta, Mar. 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.eacl-demo.16. URL `https://aclanthology.org/2024.eacl-demo.16/`.

Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang. Retrieval-augmented generation for large language models: A survey, 2023. URL `https://arxiv.org/abs/2312.10997`. arXiv preprint.

N. Guha, J. Nyarko, D. Ho, C. Ré, A. Chilton, A. K, A. Chohlas-Wood, A. Peters, B. Waldon, D. Rockmore, D. Zambrano, D. Talisman, E. Hoque, F. Surani, F. Fagan, G. Sarfaty, G. Dickinson, H. Porat, J. Hegland, J. Wu, J. Nudell, J. Niklaus, J. Nay, J. Choi, K. Tobia, M. Hagan, M. Ma, M. Livermore, N. Rasumov-Rahe, N. Holzenberger, N. Kolt, P. Henderson, S. Rehaag, S. Goel, S. Gao, S. Williams, S. Gandhi, T. Zur, V. Iyer, and Z. Li. Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 44123–44279. Curran Associates, Inc., 2023. URL `https://proceedings.neurips.cc/paper_files/paper/2023/file/89e44582fd28ddfea1ea4dcb0ebbf4b0-Paper-Datasets_and_Benchmarks.pdf`.

V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih. Dense passage retrieval for open-domain question answering. In B. Webber, T. Cohn, Y. He, and Y. Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online, Nov. 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.550. URL `https://aclanthology.org/2020.emnlp-main.550/`.

P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc., 2020. URL `https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf`.

V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, and D. E. Ho. Hallucination-free? assessing the reliability of leading ai legal research tools. *Journal of Empirical Legal Studies*, 22(2):216–242, 2025. doi: 10.1111/jels.12413. URL `https://onlinelibrary.wiley.com/doi/abs/10.1111/jels.12413`.

A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 9802–9822, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.546. URL `https://aclanthology.org/2023.acl-long.546/`.

S. Min, K. Krishna, X. Lyu, M. Lewis, W.-t. Yih, P. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In H. Bouamor, J. Pino, and K. Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore, Dec. 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.741. URL https://aclanthology.org/2023.emnlp-main.741/.

J. Niklaus, V. Matoshi, P. Rani, A. Galassi, M. Stürmer, and I. Chalkidis. LEXTREME: A multi-lingual and multi-task benchmark for the legal domain. In H. Bouamor, J. Pino, and K. Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3016–3054, Singapore, Dec. 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.200. URL https://aclanthology.org/2023.findings-emnlp.200/.

C. E. Öztürk, Ş. B. Özçelik, and A. Koç. A transformer-based prior legal case retrieval method. In *2023 31st Signal Processing and Communications Applications Conference (SIU)*, pages 1–4, 2023. doi: 10.1109/SIU59756.2023.10223938.

A. Petrov, E. La Malfa, P. Torr, and A. Bibi. Language model tokenizers introduce unfairness between languages. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 36963–36990. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/74bb24dca8334adce292883b4b651eda-Paper-Conference.pdf.

S. Robertson and H. Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019.

J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In K. Duh, H. Gomez, and S. Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.20. URL https://aclanthology.org/2024.naacl-long.20/.

S. Schweter. BERTurk - BERT models for Turkish, Apr. 2020.

W. Shi, X. Han, M. Lewis, Y. Tsvetkov, L. Zettlemoyer, and W.-t. Yih. Trusting your evidence: Hallucinate less with context-aware decoding. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), Short Papers*, pages 783–791. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.naacl-short.69. URL https://aclanthology.org/2024.naacl-short.69/.

C. Toraman, E. H. Yilmaz, F. Şahinuç, and O. Ozcelik. Impact of tokenization on language models: An analysis for turkish. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.*, 22

(4), Mar. 2023. ISSN 2375-4699. doi: 10.1145/3578707. URL https://doi.org/10.1145/3578707.

G. Uludoğan, Z. Balal, F. Akkurt, M. Turker, O. Gungor, and S. Üsküdarlı. TURNA: A Turkish encoder-decoder language model for enhanced understanding and generation. In L.-W. Ku, A. Martins, and V. Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10103–10117, Bangkok, Thailand, Aug. 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.600. URL https://aclanthology.org/2024.findings-acl.600/.

P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, and Z. Sui. Large language models are not fair evaluators. In L.-W. Ku, A. Martins, and V. Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, Bangkok, Thailand, Aug. 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.511. URL https://aclanthology.org/2024.acl-long.511/.

L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. Gonzalez, and I. Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/91f18a1287b398d378ef22505bf41832-Paper-Datasets_and_Benchmarks.pdf.